

Project Proposal

ClinicBuddy AI: A Multi-Agent Healthcare Assistant for Patient Monitoring and Clinical Summarization

1. Problem Statement & Motivation

- In the modern healthcare industry, both patients and clinicians are facing some challenges, such as managing and interpreting health data. Patients have numerous health metrics measured and recorded daily, for example, blood pressure, heart rate, and weight, but they don't have enough medical knowledge to understand the significance of these values or whether it shows potential health risks. It is leading to the result that the health data is often underutilized and misinterpreted.
- At the same time, healthcare organizations are facing a shortage of health data analysts and medical informatics specialists. One doctor must manually review a ton of patient data within limited timeframes, which creates inefficiencies and increases the chance of missing important information. Existing healthcare systems now mostly work on a primary function as a passive data storage platform, not as an intelligent decision support tool.
- ClinicBuddy AI is focused on how to solve these challenges by giving a multi-agent simulated healthcare assistant that supports basic patient monitoring and clinical summarization. This system proves how agent-based artificial intelligence can take a role as a virtual health data analyst, which automatically analyzes routine tasks and improves access to meaningful health information for both patients and physicians.

2. Target Users

- ClinicBuddy AI is focused on two primary user groups:
 - a. **Patients:** Individuals who can get access and track their health metrics such as blood pressure, heart rate, and weight. These users will receive simplified explanations of their data to improve health literacy and their awareness of the situation that they have.
 - b. **Clinicians:** Healthcare providers who must review patients' histories and require concise summaries and they need a short summary and risk indicators to support decision-making.
- ⇒ *This system does not provide medical diagnoses or treatment recommendations.*

3. Why is an Agentic AI Solution Appropriate?

- ClinicBuddy AI is using agentic AI approach because a single AI system cannot handle several related tasks that healthcare data interpretations are required to. These tasks include validating and storing patients' data, analyzing trends over time, identifying

abnormal values, generating clinical-style summaries and making sure that outputs are safe and appropriate.

- A multi-agent lets different tasks being managed by different special agents with specific roles. That will help to make the system more organized and transparent, showing step-by-step reasoning, and giving out the reflection of real-world healthcare settings in which multiple professionals collaborate to manage patients' information.

4. Use Cases and Scenarios

a. Scenario 1: Patient Monitoring

- A patient log into ClinicBuddy AI and views their daily vital signs, including blood pressure, heart rate, and weight. The system will automatically analyze the data, show trends over time, and provide easy to understand explanations of the results.

b. Scenario 2: Clinical Review

- A clinician selects a patient profile and views a dashboard showing trends overtime. The system is generating a full informative clinical summary highlighting recent changes and risk flags.

c. Scenario 3: Safety Verification

- If a patient asks health-related questions, such as “Is my heart rare in the dangerous condition?”, the system just provide educational information while avoiding medical diagnosis treatment advice to ensure the safety of the agent.

5. Initial Systems Design

- ClinicBuddy AI consists of five specialized agents:

Agent	Role
1/ Data Intake Agent	Confirms and stores patients' health data
2/ Trend Analysis Agent	Analyze the patient's historical vital signs and generate trends graph to show if the condition is improving, stable or worsening based on recent data.
3/ Risk Flags Agent	Identifies abnormal values using rule-based logic.
4/ Doctor Summary Agent	Generate clinician-facing summaries.
5/ Safety Agent	Reviews outputs for unsafe medical advice.

Tool and Data Sources:

- Python functions for data validation and analysis.
- Local database for storing and controlling patient records.
- LLM API for natural language generation.

- Streamlit for the web-based front-end interface and build-in charts.

6. Feasibility and Risks

a. Potential Challenges:

- Language models might generate inaccurate or misleading information.
- Output may lead to medical advice.
- Simulated data cannot reflect real-world complexity.

b. Mitigation Strategies:

- Restrict output to just explanations and summaries only.
- Implement a Safety agent and test all the functions to filter unsafe outputs.
- Use rule-based risk detection while developing predictive models.

7. Project Timeline

- Proposal: Define problem and system design
- Milestone I: Implement front-end interface
- Milestone II: Integrate multi-agent workflow
- Milestone III: Conduct evaluation and pilot testing
- Final Phase: Record demo video and submit final report